The Culture of Business, Finance and the Economy

Class 7 – 26 march

Dr. Juan M. del Nido

First semester 2026

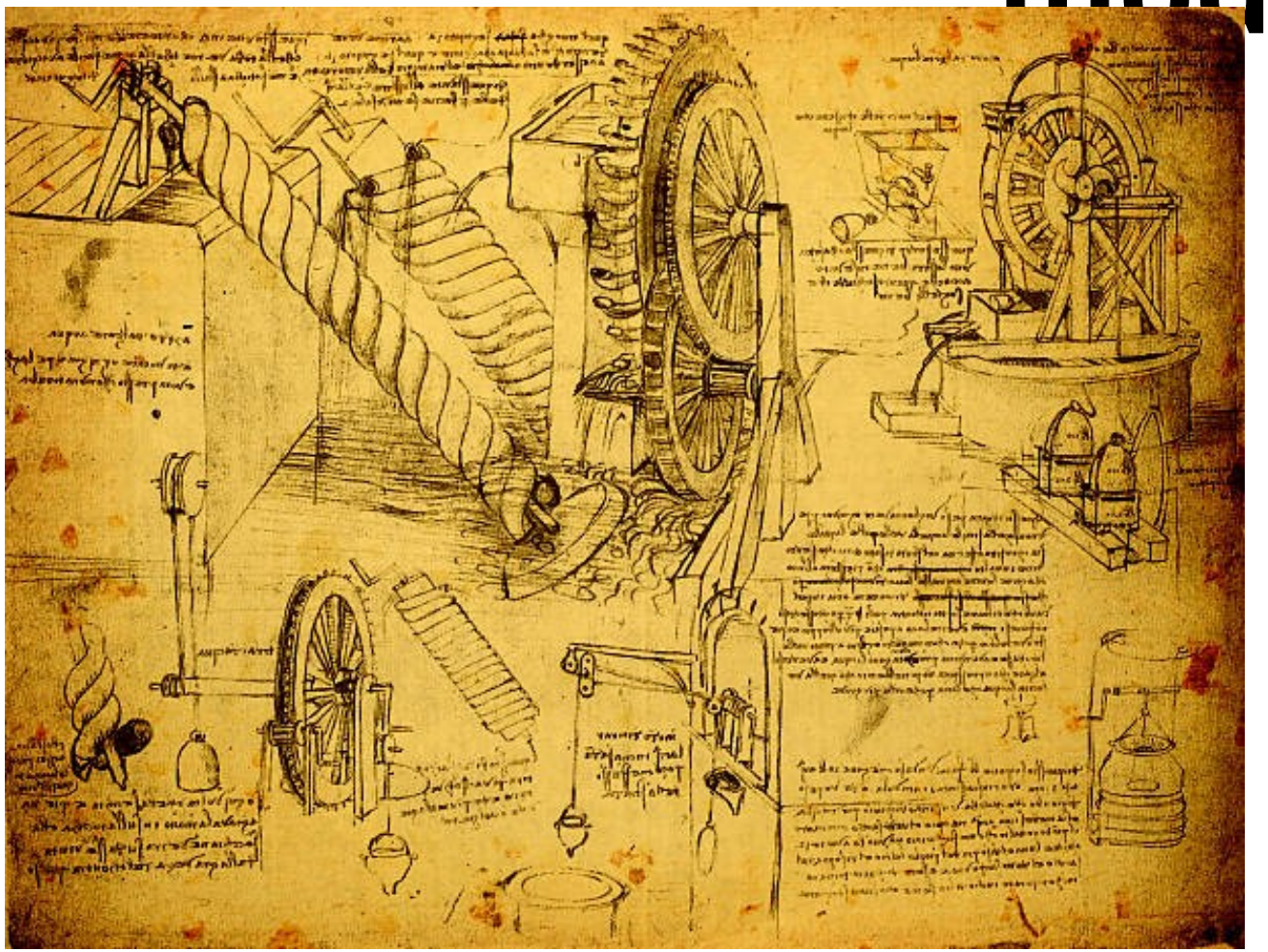
Recap

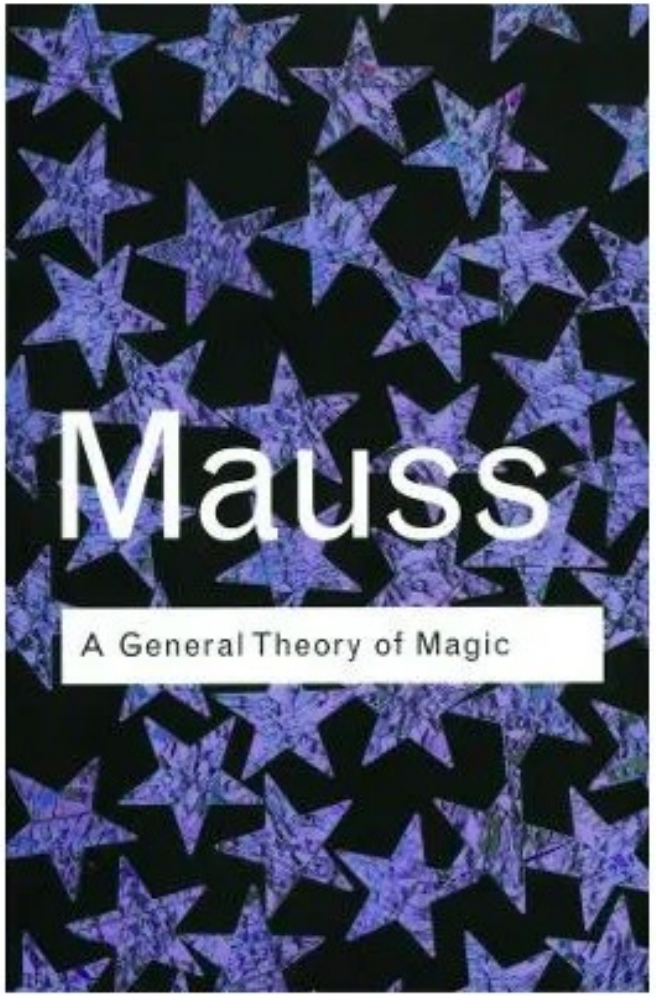
Physics and mathematics (last 3 classes)

“defying the laws of gravity”

To challenge Economics is to challenge Physics and Mathematics, which is not just ”a disagreement”, but also devious, ignorant, and even a moral wrong, against the natural order of things

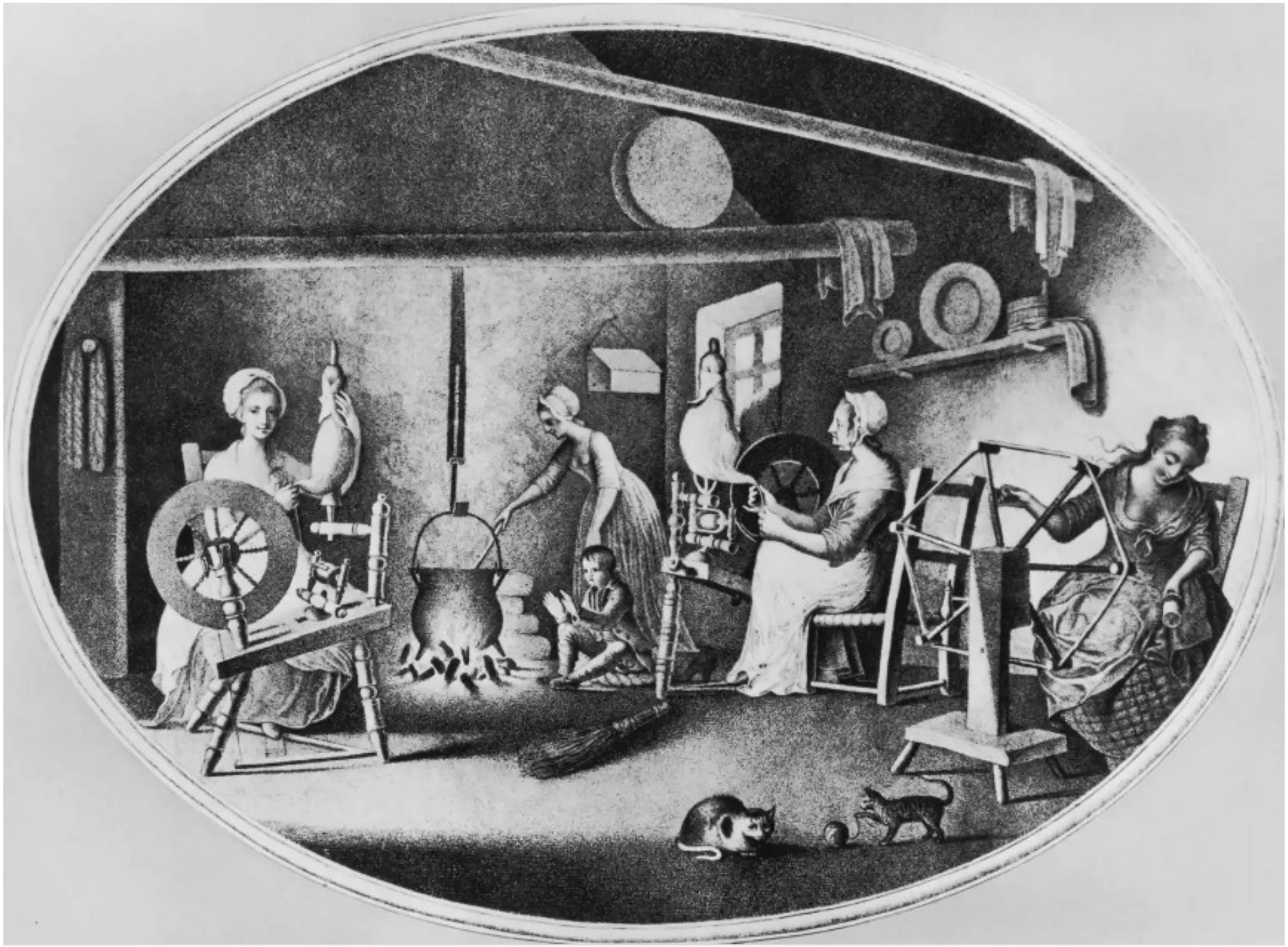
Today
Thinking with, and through, models





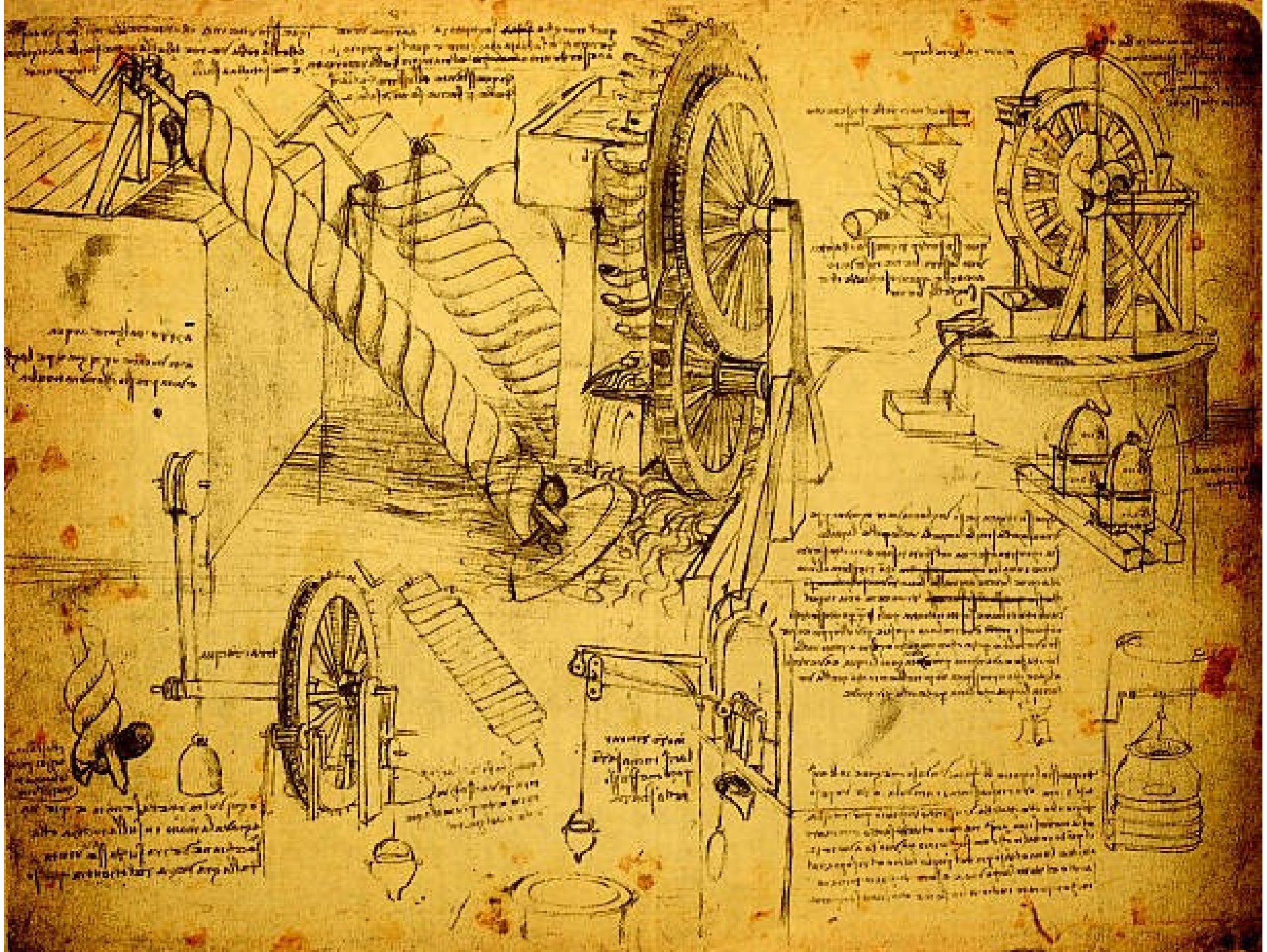
What is a model in business, finance and the economy?

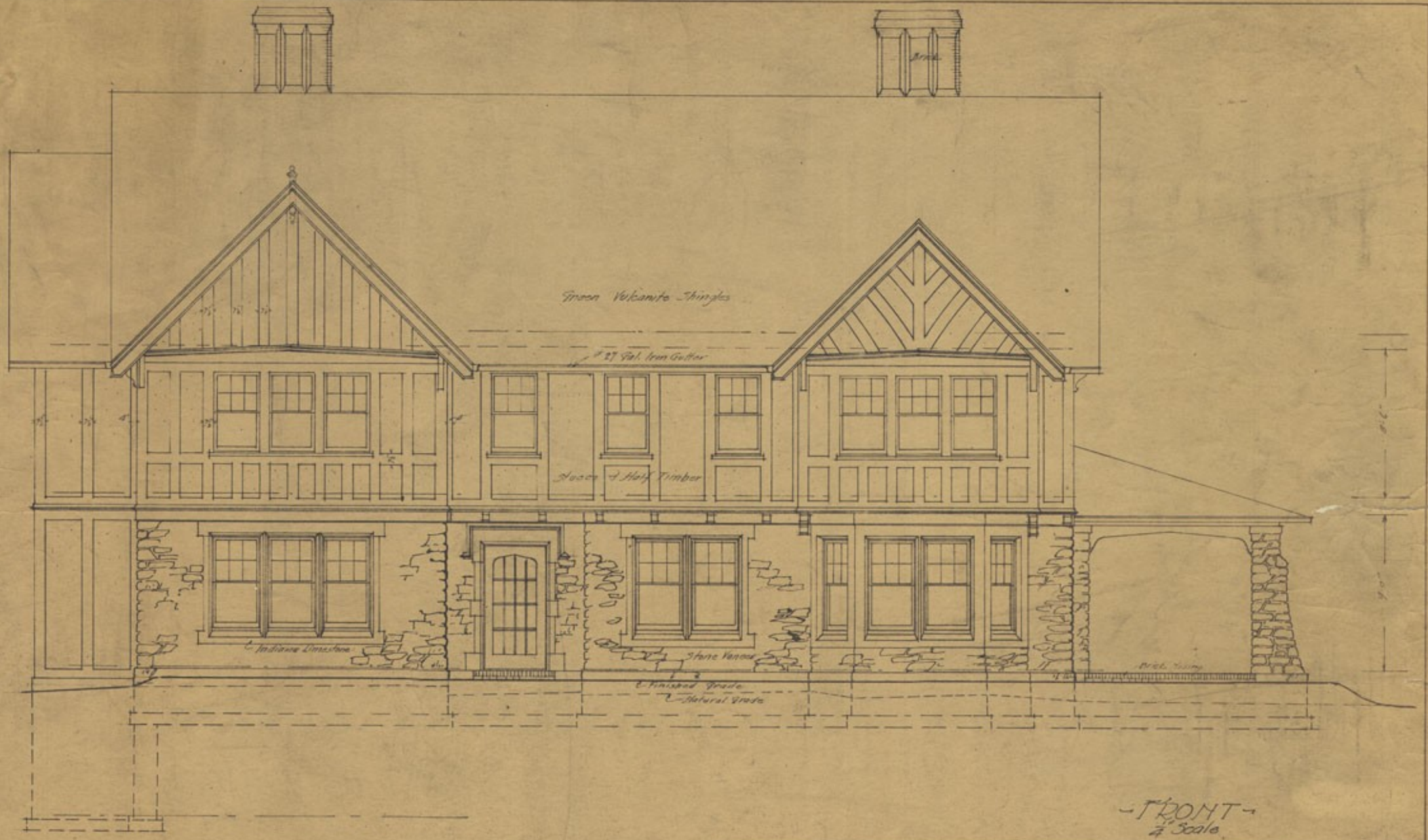
HIC TRAHVNT: NAV



IVNIVS

Guelches I.
Duquedec
Gismel
Cahuashets
Senchueta
Castro
Chiloe
R. de S. Josef
R. Chico de S. Josef
Cozinha
Secumelel
R. de los Camarone
Chutillan - Cunis I.
I. del Rey
Archip. de los Chanes
Chonos I.
Bay de S. Jorge
Vole de S. Clemente
Halbinsel de tres Montes
S. Coluguape
Monter
Meerb. v. Penas
S. Cheleuco
R. Galleno
Quisahet
Guayaneco
R. de las Cauenas
Arquetes I.
R. Desonde
Isle N. Senora
Apostelos





A RESIDENCE for JUDGE J. G. ADAMS at BILTMORE FOREST

Charles H. Parker
Archit. Jan 1921

SA0206.1

Late 1750s, François Quesnay, court of Louis XV

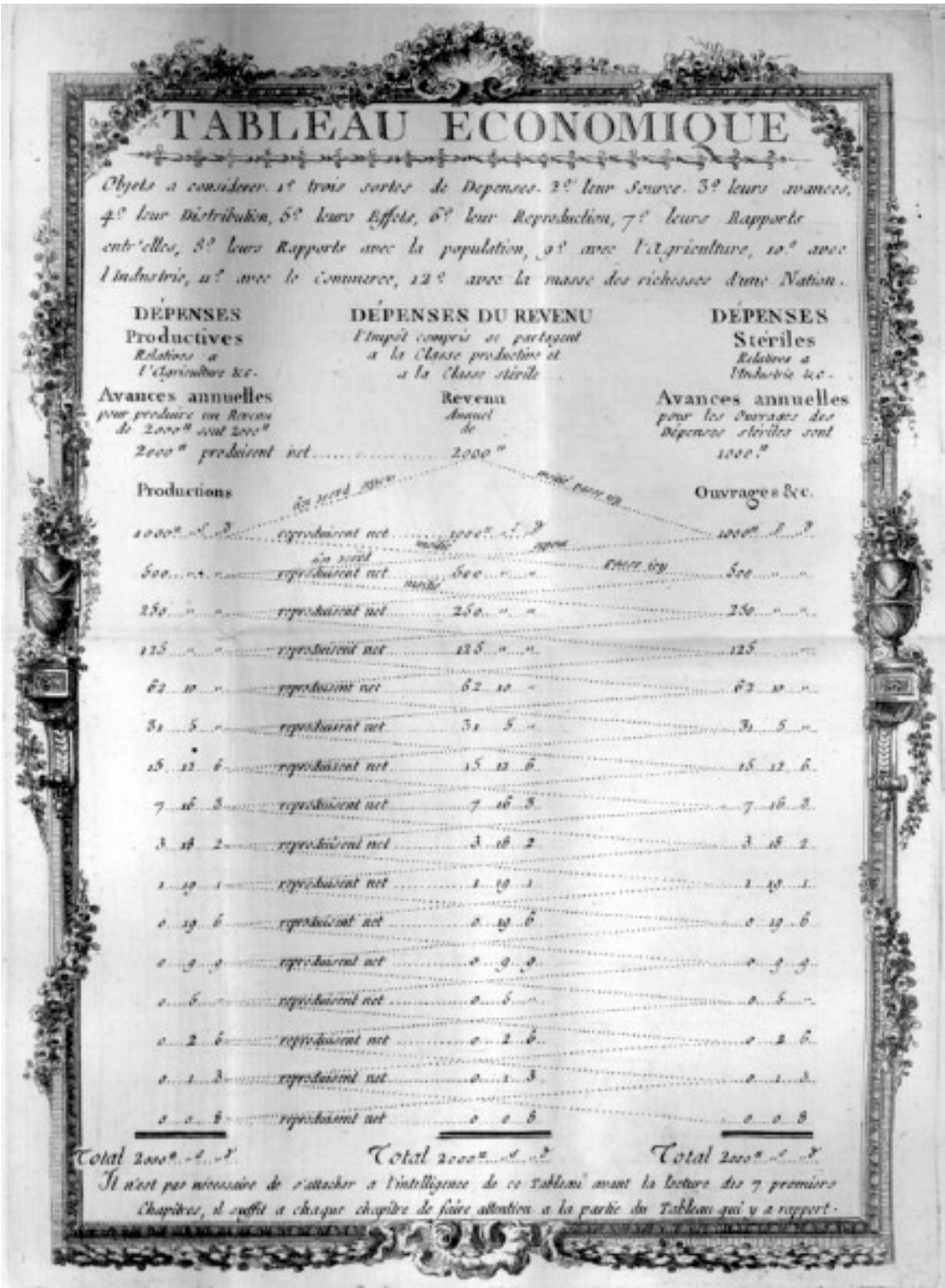


TABLEAU ECONOMIQUE

Objets a considerer. 1° trois sortes de Depenses. 2° leur Source. 3° leurs avances, 4° leur Distribution, 5° leurs Effets, 6° leur Reproduction, 7° leurs Rapports entr'elles, 8° leurs Rapports avec la population, 9° avec l'Agriculture, 10° avec l'Industrie, 11° avec le Commerce, 12° avec la masse des richesses d'une Nation.

DÉPENSES Productives Relatives a l'Agriculture &c.		DÉPENSES DU REVENU l'impôt compris se partagent a la Classe productive et a la Classe stérile		DÉPENSES Stériles Relatives a l'Industrie &c.
Avances annuelles pour produire un Revenu de 2000# sont 2000#		Revenu Annuel de		Avances annuelles pour les Ouvrages des Dépenses stériles sont 1000#
2000# produisent net		2000#		
Productions				Ouvrages &c.
1000#	reproduisent net	1000#		1000#
500	reproduisent net	500		500
250	reproduisent net	250		250
125	reproduisent net	125		125
62 10	reproduisent net	62 10		62 10
31 5	reproduisent net	31 5		31 5
15 12 6	reproduisent net	15 12 6		15 12 6
7 16 3	reproduisent net	7 16 3		7 16 3
3 18 2	reproduisent net	3 18 2		3 18 2
1 19 1	reproduisent net	1 19 1		1 19 1
0 19 6	reproduisent net	0 19 6		0 19 6
0 9 9	reproduisent net	0 9 9		0 9 9
0 5	reproduisent net	0 5		0 5
0 2 6	reproduisent net	0 2 6		0 2 6
0 1 3	reproduisent net	0 1 3		0 1 3
0 0 8	reproduisent net	0 0 8		0 0 8
Total 2000#		Total 2000#		Total 2000#

Il n'est pas nécessaire de s'attacher a l'intelligence de ce Tableau avant la lecture des 7 premiers Chapitres, il suffit a chaque chapitre de faire attention a la partie du Tableau qui y a rapport.

TABLEAU ECONOMIQUE

Objets à considérer. 1° trois sortes de Dépenses. 2° leur Source. 3° leurs avances, 4° leur Distribution, 5° leurs Effets, 6° leur Reproduction, 7° leurs Rapports entr'elles, 8° leurs Rapports avec la population, 9° avec l'Agriculture, 10° avec l'Industrie, 11° avec le Commerce, 12° avec la masse des richesses d'une Nation.

DÉPENSES Productives *Relatives à l'Agriculture &c.*	DÉPENSES DU REVENU *l'Impôt compris se partageant à la Classe productive et à la Classe stérile*	DÉPENSES Stériles *Relatives à l'Industrie &c.*
Avances annuelles *pour produire un Revenu de 2000# sont 2000#*	Revenu *Annuel de*	Avances annuelles *pour les Ouvrages des Dépenses stériles sont*
2000# *produisent net*	2000#	1000#
Productions		Ouvrages &c.
1000# ...?... *reproduisent net*	1000# ...?...	1000# ...?...
500 ... *reproduisent net*	500 ...	500 ...
250 ... *reproduisent net*	250 ...	250 ...
125 ... *reproduisent net*	125 ...	125 ...

A cross between a table, a matrix, a portrait of the French economy. It shows classes of people (farmers, manufacturers, and landowners) and the zig-zag pattern of lines between them indicate goods or money circulating between them.

It was a research object, to conduct numerical exercises to explore how, after agricultural investment, surplus would circulate.

The great-grandfather of models

Models depict some aspect of the economy in a schematic, simplified, way. They are never simply a passive portrait of the economy and they don’t just work through analogy; their inner workings let us understand the logics that govern a particular phenomenon or process.

It is this possibility for <u>managed manipulation</u> that turns such pictures, diagrams, equations, prototypes, etc., into models in the culture of business, finance and the economy.

mathematical, statistical, graphical, diagrammatic, and even physical tools and objects (remember the MONIAC) that one can manipulate and interact with…

…so as to reason with and think through them.

A very specific kind of scientific object

JANUARY
FEBRUARY
MARCH

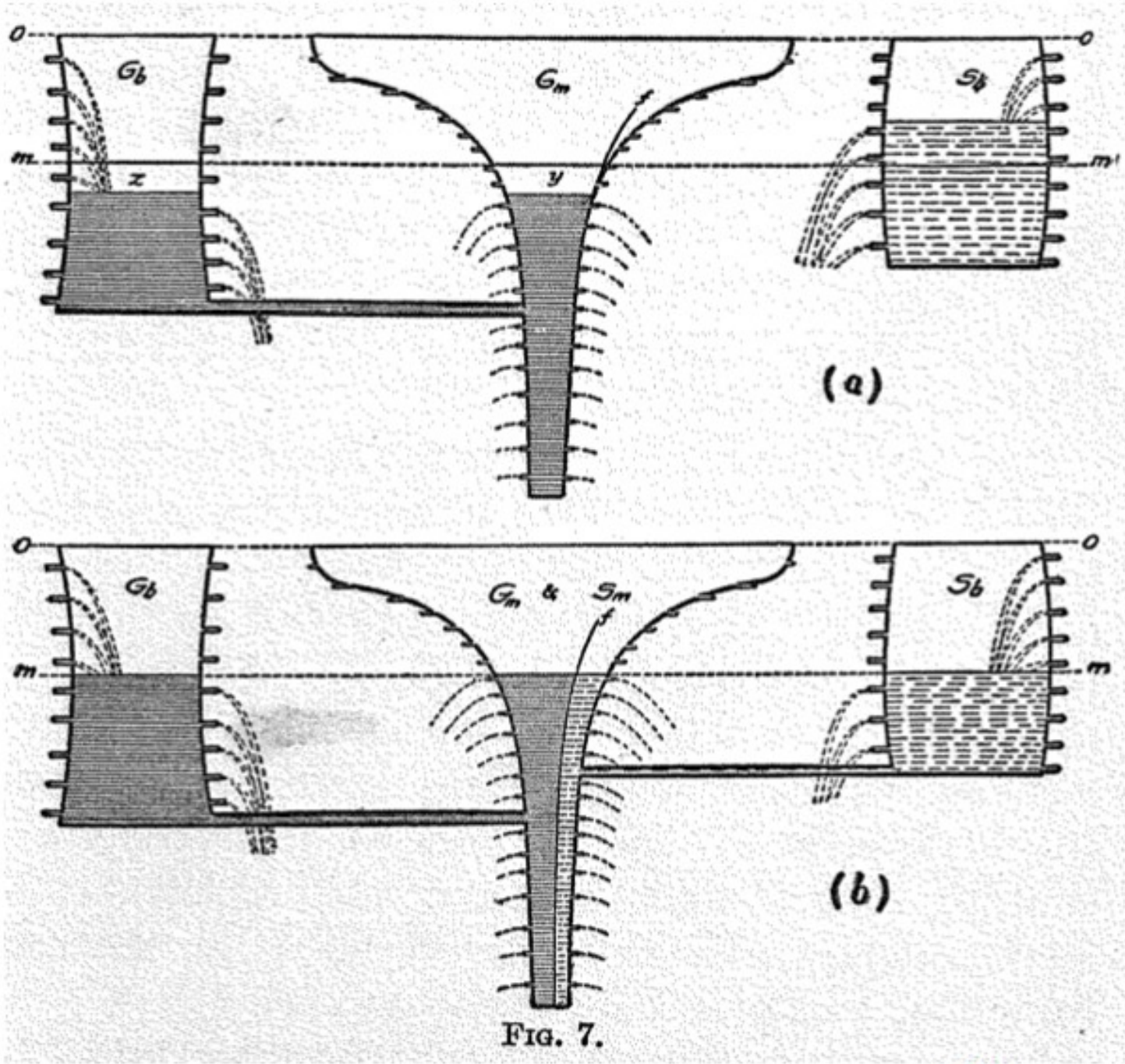


Figure 2: Fisher making use of hydraulics to model the purchasing power of money under a system of bimetallism [21]

INVESTMENT FUNDS
INVESTMENT
GOVERNMENT SPENDING
DOMESTIC EXPENDITURE
WHITE ELLERTON
NATIONAL INCOME

They must be small enough in scale to be fiddled with to enquire – indirectly – into the workings of those aspects of the economy depicted. They also need to allow to further theory and to hypothesise by enabling experimentation with bits we do not yet fully know.

This opens questions about accuracy, precision, relevance and the kinds of relations the model seeks to transpose…

…but it opens them in a peculiar way (coming soon)

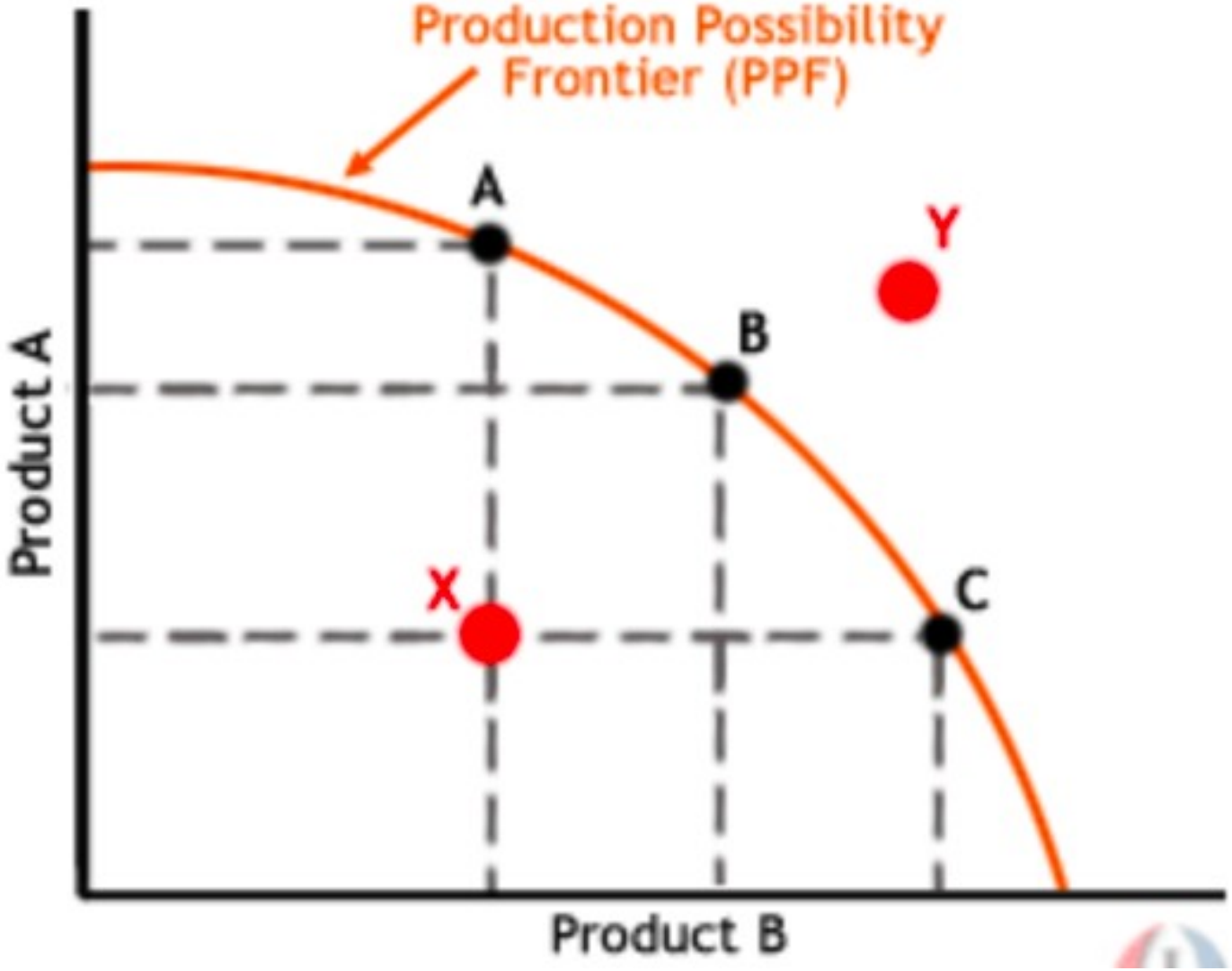
2 kinds of rules

Formal rules

(mathematics – geometry, algebra, arithmetic – physics, etc.)

Substance rules

(economic principles)


Production Possibility
Frontier (PPF)
A
B
C
X
Y
Product A
Product B

Models have to be <u>bounded</u>

The end point has to be clear. They have to be self contained, which is to say whatever they include has to be enough to know that which they claim to render knowable.

What is there is what you need to know; it is enough to know

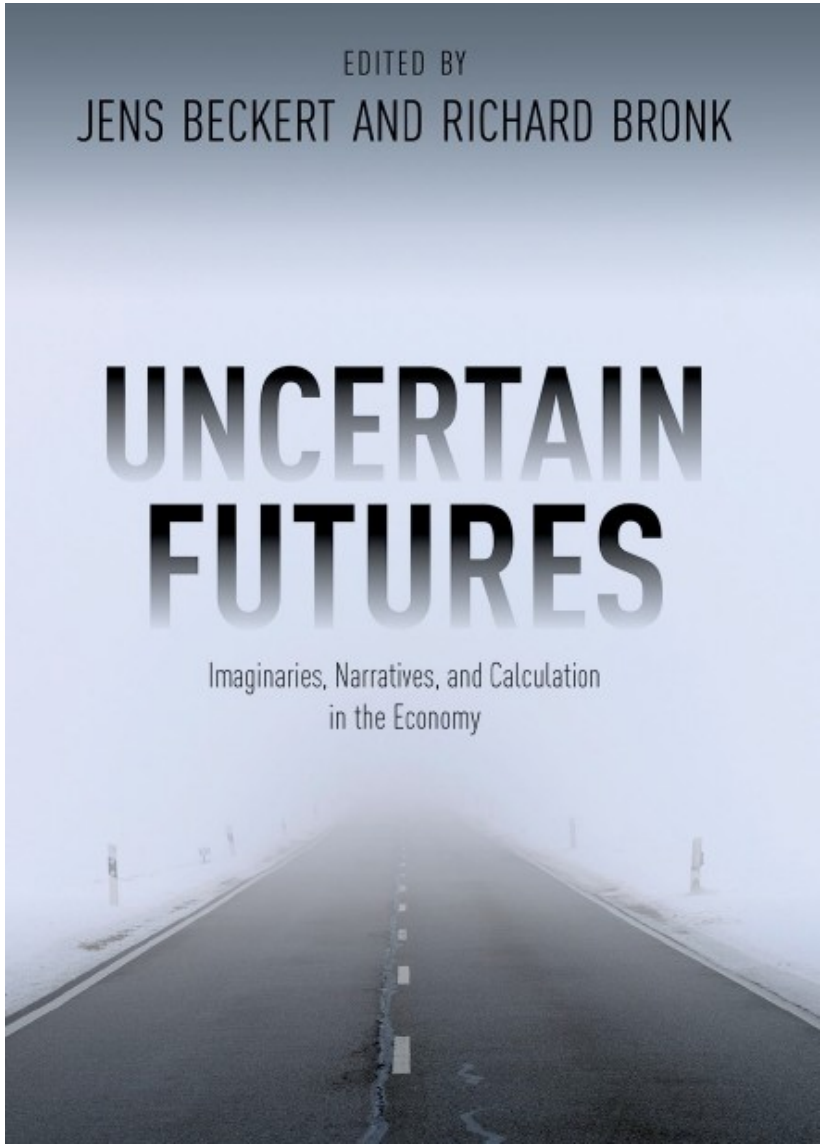
Why do we care?

6

Escaping the Reality Test

How Macroeconomic Forecasters Deal With 'Errors'

Olivier Pilmis



Break

Why do we care?

6

Escaping the Reality Test

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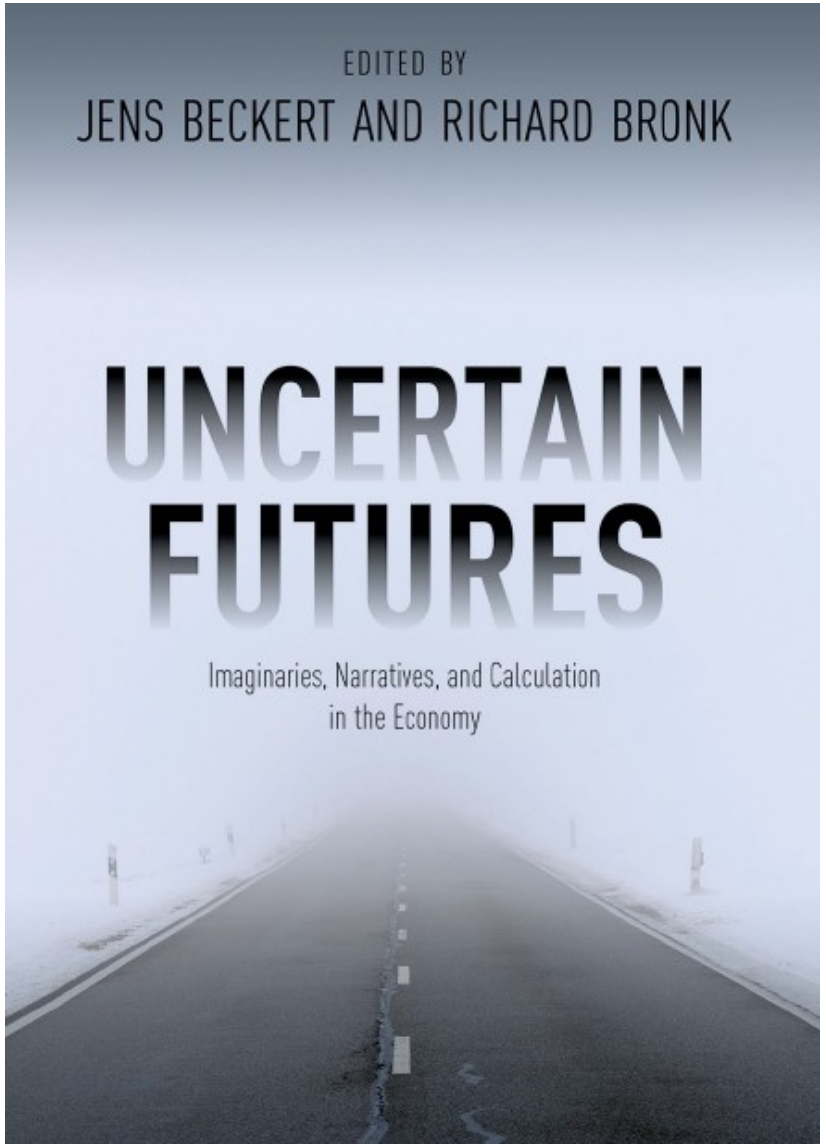

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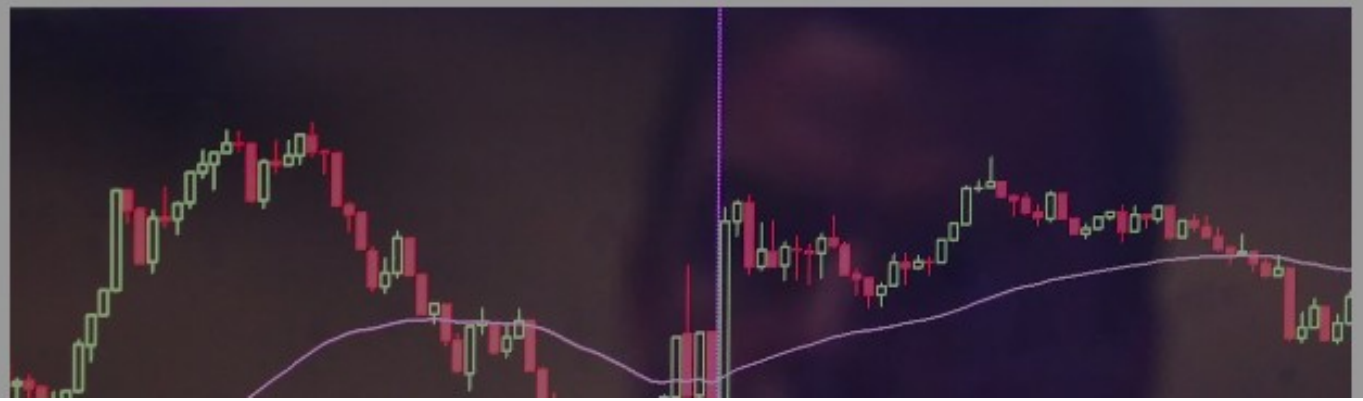
Markets
Merryn Talks Money

The Problem With Financial Forecasts Is They're Usually Wrong

On this episode of *Merryn Talks Money*, we discuss why we use them anyway.



The Global Financial Crisis – Why Didn't Anybody Notice?

On 17 June 2009, a group of leading academics, economics journalists, politicians, civil servants, and other practitioners met at the British Academy for a round-table discussion of the current financial crisis. The question under discussion in this British Academy Forum had been framed by Her Majesty The Queen on a visit to the London School of Economics in November 2008, when she had asked: if these things were so large, how come everyone missed them? A purpose of the Forum was to provide the basis of an 'unofficial command paper' that attempted to answer this question. The discussion inevitably ranged more widely – touching on the social fall-out of the crisis, and including a plea for a greater emphasis on the teaching of economic history in universities. But it was with The Queen's question in mind that the two convenors of the meeting, **Professor Tim Besley FBA** *and* **Professor Peter Hennessy FBA,** *subsequently drafted a letter summarising the discussion: it was sent to Buckingham Palace on 22 July.*

Figure 1. *Queen Elizabeth II is shown a display about the credit crunch by Professor Luis Garicano, during a visit to the London School of Economics and Political Science on 5 November 2008. Photo: Wigglesworth/PA Wire URN:6524464 (Press Association via AP Images).*

MADAM,

When Your Majesty visited the London School of Economics last November, you quite rightly asked: why had nobody noticed that the credit crunch was on its way? The British Academy convened a forum on 17 June 2009 to debate your question, with contributions from a range of experts from business, the City, its regulators, academia, and government. This letter summarises the views of the participants and the factors that they cited in our discussion, and we hope that it offers an answer to your question.

Many people did foresee the crisis. However, the exact form that it would take and the timing of its onset and ferocity were foreseen by nobody. What matters in such circumstances is not just to predict the nature

Business Management

Business Forecasts Are Reliably Wrong — Yet Still Valuable

by Martin Reeves, Suvasini Ramaswamy and Annelies O'Dea

March 8, 2022

ARCHIVE

Why Economists' Predictions Are Usually Wrong

They almost always fail to foresee a recession before it happens. But there are ways they can improve their insights.

October 29, 2015 • **Liz Farmer**

Most Read

1 The Key Ingredients of Success

How do economics, finance and business forecasters get away with being wrong so often, or with “failing the reality test”?

1 – the market is immense

2 – collective ignorance

3 – emphasis on narrative

4 – emphasis on procedure and compliance

1

The market is unknowable because it is immense. By definition, models focus on a selection of variables when they make predictions: an exhaustive model is impossible.

When something unpredictable happens – that is, something that those variables, and that how we handled them, did not foresee – we tend to treat it as an external, unpredictable shock, 'exogenous' to the model - and therefore as legitimate excuses for forecasting 'errors'.

This unknowability is a <u>resource</u> for economists. If a prediction fails, economists can claim it wasn't their fault: markets are just too immense to fit a model bounded by definition. It's not that they were wrong, or even that the model was wrong –it just didn't account for x, y or z.

This leads to the apparently paradoxical conclusion that models should be improved rather than withdrawn …

…thus perpetuating our reliance on models and on those who make and handle them.

2

CATO INSTITUTE

Where Forecasting Went Right

JANUARY 5, 2006 • COMMENTARY

By Alan Reynolds

SHARE

"Forecasts tend to cluster close together, because it's risky to go out on a limb. The expert judgment of 56 economic forecasters in *The Wall Street Journal* is not something that can be brushed off that easily"

Collective ignorance offers an ‘alibi’ that deflects individual accountability: the very fact that no other expert knew about or understood what was coming “proves” it was impossible to know.

Collective ignorance does not make the reality test ineffective as such but it makes economists immune to its verdict as they are not accountable for not having predicted what was unpredictable: forecasts were ‘wrong’ but no error was made (and this wrongness thus cannot challenge the fundamentals of financial or economic knowledge or the authority of economists)

3

Economists often present narratives as the ‘real’ forecasts, those that should be paid attention to rather than ‘point’ forecasts. In this perspective, forecasting has more to do with storytelling than with calculation. Whereas it seems easier to assess the accuracy of forecasts, and thus their ‘errors’, with reference to calculated numbers, to economists themselves the quality of storytelling is a more appropriate test of forecasts. To fellow forecasters, a ‘good’, or ‘right’, forecast ‘correctly’ describes macroeconomic progression in such a way that the ‘story’ matches the coming string of events, even if the anticipated figures are ‘wrong’. Thus forecasting is not subject to simple falsification because it can accommodate “being right for the wrong reasons” and viceversa.

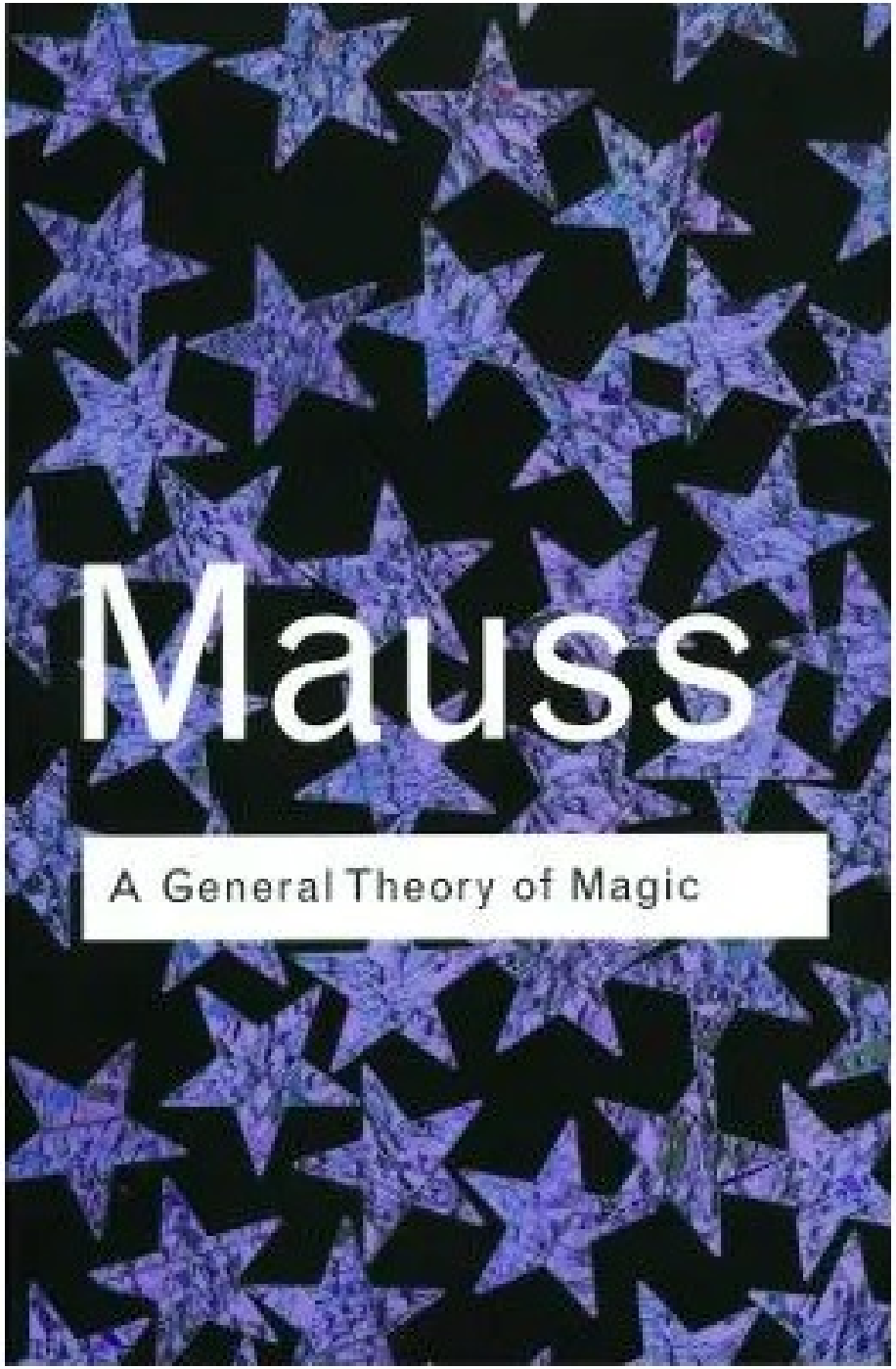
4

Economists who go by the book are exonerated of forecasting ‘errors’: they are not to be held responsible for ‘mistakes’ since none have been made. This approach to ‘error’ relates it exclusively to the degree of compliance with internally defined standard practices and formal rules. This emphasis on procedures seem connected to their claim that their professional identity is first and foremost that of economists who *shall be judged according to the narratives they set up rather than the calculations they end up with*. Protection from blame is obtained by proving that the ritual of forecasting has been followed: insofar as the process has been correctly handled, there is no point in criticizing its outcomes

So Pilmis concludes…

Modelling is a *belief-based* practice (not a result-based practice)…

… and a lot like magic


Mauss
A General Theory of Magic

the authority of magic, magicians and shamans – like that of models and economists – originates in collective ideas about how important they are, so much so that beliefs and confidence in magic are a priori rather than evidence-based. A priori confidence in magic, and in economics, arises from the collective need to find explanations to phenomena that would otherwise remain inexplicable. Thus, magical beliefs and economics stand firm when challenged.

models are to the culture of business, finance and the economy as magical tools are to magic: they are used "for their own sake by virtue of their real or imaginary properties"

The narrative – the spell – matters! You need to get the spell right. Like magical rituals, economic forecasting implies that ‘the preparation of [the] materials, the ritual ingredients, is a long and finicky business’. The emphasis on procedure – following steps in order, etc. is crucial. If the outcome is not what you expected, it’s not because modelling – or magic – in themselves are unfit for purpose, but because you didn’t cast the spell properly, or you missed a step, or because you are, after all, trying to domesticate the unknown, the universe or the market, and there may be something beyond that you missed.

As with magic, belief in forecasting remains to a certain extent immune to contradictory experiences because it may be argued that the ritual was not properly executed.

The conditions required for magic to be successful are so many that ‘to such an extent that it seems to be providing itself with loopholes, and often successfully’, so that ‘it is natural for a magician to take refuge behind questions of procedures and technicalities, to protect himself in case of failure of magical prowess’ (Mauss).

Replace “magic” for “models” and “magician” for “economist”

The model is a self-validating buffer

A means for economics and finance to know the world, self-validate that knowledge, and deflect criticism of how that knowledge works

Outside the model?

Example: Taxis in Buenos Aires

Models as tools we can think with…

…models as tools we can’t think without